Manu Jayadharan, PhD

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Professional Summary

Applied mathematician and computational scientist, currently a postdoctoral fellow in applied mathematics at Northwestern University with two years of industrial experience as a quantitative analyst, developing reliable data-driven methods for scientific systems. My work includes equation discovery for differential and differential-algebraic systems, identifiability diagnostics, multiple-shooting methods for stiff inverse problems, hybrid mechanistic-neural models, and finite element and domain-decomposition solvers for multiphysics PDEs. I am also interested in responsible agentic AI for scientific computing: using AI agents for model construction, code generation, and computational experimentation while retaining numerical validation, reproducibility, and human accountability. My independent-research record includes two student-led manuscripts as senior author and first-hand experience contributing to the development and writing of research-grant proposals. Two years in the Wall Street financial industry as a Quantitative Analyst on the Global Credit team at Citigroup, a global investment bank, gave me experience with production software, large-scale model validation, and the practical consequences of model risk.

Research Interests

Data science; scientific machine learning; data-driven equation discovery; differential and differential-algebraic systems; sparse optimization; inverse problems; identifiability and uncertainty; responsible agentic AI for scientific computing; interpretable AI for science; machine learning and optimization; numerical analysis; finite-element methods; multiscale and mixed methods; domain decomposition; computational and multiphysics PDE; numerical linear algebra; HPC-aware computational science; electrochemical and biological systems; scientific software.

Education

University of Pittsburgh, Pittsburgh, PA, USA — Aug 2016 – Aug 2021
Ph.D. in **Mathematics** (area: numerical analysis, scientific computing, and HPC for PDEs). Advisor: Dr. Ivan Yotov.
Dissertation: *Domain Decomposition and Time-Splitting Methods for the Biot System of Poroelasticity* (2021).

IISER Mohali, Mohali, India — Aug 2011 – May 2016
BS-MS Dual Degree in **Mathematics** (integrated undergraduate and postgraduate degree). Full DST INSPIRE fellowship; top rank holder, Department of Mathematics.
MS Thesis: *Study of Cauchy's Basic Equations and Convex Functions* (2016).

Academic Positions

Northwestern University, Evanston, IL, USA — Sept 2023 – Present
Postdoctoral Fellow, Department of Engineering Sciences and Applied Mathematics (PI: Dr. Niall Mangan). Affiliated with the National Institute for Theory and Mathematics in Biology (NITMB), an NSF–Simons-funded center for mathematical biology, and the Trienens Institute for Sustainability and Energy.

- Developed sparse-regression methods for discovering differential and differential-algebraic equations from noisy data, motivated by applications in electrochemistry, systems biology, chemical engineering, selected power-system examples, and battery-related modeling.
- Diagnosed ill-conditioning and structural unidentifiability in dictionary-based model discovery, connecting the failure modes to inverse-problem theory and developing QR-based stabilization.

- Developed a multiple-shooting framework with guess propagation for stiff inverse problems in model discovery.
- Led the design and open-source release of `dae-finder`, a model-agnostic Python package for discovering differential-algebraic equations from noisy data.
- Built Galerkin weak-form finite-element solvers for Poisson–Nernst–Planck electrochemical transport at the Trienens Institute.
- Lead an initiative on rigorous, responsible, and resource-conscious uses of agentic AI in scientific computing; it has led to a fully funded graduate course, *Agentic AI for Scientific Computing*, to be offered by Northwestern Applied Mathematics in Spring 2027.
- Assisted with the technical development and writing of grant proposals on fast solvers for multiphysics PDEs and data-driven PDE discovery; contributed to periodic research reporting for institute and grant stakeholders.

Industry Experience

Citigroup Inc., New York, NY, USA May 2021 – Aug 2023
Quantitative Analyst, AVP, Global Credit Quantitative Analysis. FINRA-licensed.

- Contributed to large-scale quantitative libraries for pricing financial derivatives and computing risk and sensitivity measures used by Citi's trading desks worldwide.
- Implemented new risk metrics; supported model validation and deployment within a complex legacy production codebase.
- Worked closely with traders and risk managers; the asset-agnostic design of Citi's pricing libraries later informed the model-agnostic architecture of `dae-finder`.

Publications

Author-order flags: **bold** highlights my name; † = senior author; first-author and senior-author roles are stated explicitly per entry.

Refereed Journal Articles, Published / Accepted

1. **M. Jayadharan**, C. Catlett, A. N. Montanari, N. M. Mangan, "SODAs: Sparse optimization for the discovery of differential and algebraic equations," *Proceedings of the Royal Society A*, 2026. DOI: 10.1098/rspa.2025.0201. [**First author.**]
2. **M. Jayadharan**, I. Yotov, "Multiscale mortar mixed finite element methods for the Biot system of poroelasticity," *Computer Methods in Applied Mechanics and Engineering* (CMAME), Vol. 435, Art. 117597, 2025. [**First author.**]
3. **M. Jayadharan**, M. Kern, M. Vohralík, I. Yotov, "A space-time multiscale mortar mixed finite element method for parabolic equations," *SIAM Journal on Numerical Analysis*, 2023. [**First author.**]
4. **M. Jayadharan**, E. Khattatov, I. Yotov, "Domain decomposition and partitioning methods for mixed finite element discretizations of the Biot system of poroelasticity," *Computational Geosciences*, 25, 1919–1938, 2021. [**First author.**]

Under Revision / Preprint

5. Y. Feng, N. M. Mangan, **M. Jayadharan**†, "Ill-conditioning in dictionary-based dynamic-equation learning: A systems biology case study," arXiv:2603.11330, 2026. (Under revision, *SIAM Journal on Life Sciences*.) [**Senior author.**]

6. M. Ghumman, **M. Jayadharan**, I. Yotov, "Domain decomposition methods for the Stokes–Biot model of fluid–poroelastic structure interaction," arXiv:2606.17363, 2026. (Under revision, *Journal of Computational Physics*.)

In Preparation

7. **M. Jayadharan**, N. M. Mangan, "Stabilizing Sparse Model Identification through Truly Orthogonal Candidate Libraries," 2026. **[First author. In preparation.]**

8. **M. Jayadharan**, N. M. Mangan, "A Multi-Shooting Framework for Sparse Model Discovery with Guess Propagation," 2026. **[First author. In preparation.]**

9. H. Chen, N. M. Mangan, **M. Jayadharan**†, "Improving Extrapolation in Neural ODEs via Monomial Augmentation and Symbolic Structure Recovery," 2026. **[Senior author; in preparation.]**

† *Senior author. The Feng manuscript is under revision and I am its senior and corresponding author; the Chen manuscript is a student-led work in preparation on which I am senior author. Together, they provide evidence of an independent research and mentoring program in data-driven equation learning and scientific machine learning. No co-first-author designations apply to the publications above.*

Open-Source Scientific Software

- **DAE-FINDER** (`dae-finder` on PyPI). Model-agnostic Python package for discovering differential-algebraic equations from noisy data via sparse optimization; also a substrate for comparing related equation-discovery methods.
- **FluidLearn.** Physics-informed neural network framework for multiphysics PDEs (Python).
- **MMMFE-ST-DD.** Flow simulator using mixed finite elements with multiscale space-time subdomains (C++, MPI).
- **BiotDD and BiotDD-Heterogeneous.** deal.II/MPI domain-decomposition and multiscale mortar simulators for Biot poroelasticity (C++).
- Repositories and website: mjayadharan.github.io.

Selected Invited Talks and Conference Presentations

- **Invited talk:** *From Data to Differential Equations*, Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram, August 2025. Host: Dr. Deepak Mishra.
- **Invited talk:** *SODAs for Discovering Differential-Algebraic Equations*, SIAM Annual Meeting (AN25), remote component, July 2025.
- **Flash talk and poster:** *Stable Learning of Differential Equations: Navigating the Ill-Posedness of Data-Driven Equation Learning*, ACDC Autumn School, VU Amsterdam / Leiden University, 2025.
- **Invited talk:** *On Discovery of Differential-Algebraic Equations*, SIAM Conference on Mathematics of Data Science (MDS24), October 2024.
- **Conference talk:** *On Numerical Methods for Discovering Differential Equations*, SIAM Annual Meeting (AN24), Spokane, Washington, July 2024.
- **Invited minisymposium talk:** *Domain Decomposition and Partitioning Methods for Mixed Finite Element Discretizations of the Biot System of Poroelasticity*, SIAM Conference on Mathematical and Computational Issues in the Geosciences, Milan, June 2021.
- **Conference talk:** *Discretization Techniques for the Biot System of Poroelasticity*, Finite Element Circus, Virginia Tech, Blacksburg, Virginia, November 2019.

- **Poster:** *Cauchy Equations and Solutions*, IISER Mohali, April 2016.
- **Research talk:** *A Variable-Topology Neural-Network Simulator*, Indian Institute of Space Science and Technology, Thiruvananthapuram, July 2015.

Conference Organization

- Organizer, three-part Minisymposium on *Advances in Discovering Interpretable Models from Fully and Partially Observed Data*, bringing together 12 speakers from institutions around the world, SIAM Conference on Mathematics of Data Science (MDS26), Salt Lake City, November 2026.
- Co-organizer with Dr. Niall Mangan, NITMB workshop series on *Data-Driven Sparse Model Selection*, 2025–2026; coordinated scientific programming on sparse model discovery, identifiability, and uncertainty.
- Organizer, Minisymposium on *Numerical Methods for Sparse Optimization*, SIAM Annual Meeting (AN24), Spokane, Washington, July 2024.

Teaching Experience

- **Northwestern University**, Fall 2023. *Instructor*, Engineering Analysis 4: Differential Equations and Numerical Methods.
- **University of Pittsburgh**, Summer 2017. *Instructor*, MATH 290: Introduction to ODEs and Applications.
- **University of Pittsburgh**, 2016–2021. *Teaching Fellow / Teaching Assistant*: Calculus I–III, Business Calculus, Ordinary Differential Equations.

Planned course development: *Agentic AI for Scientific Computing* (Northwestern, Applied Mathematics): fully funded graduate course to be offered in Spring 2027 (not yet taught), focused on responsible use, validation, reproducibility, and resource-conscious workflows for AI in scientific computing.

Mentoring

Mentored undergraduate and graduate students across Northwestern, UC Berkeley, Carnegie Mellon University, IIST, and the University of Pittsburgh. This mentoring has produced two student-led manuscripts on which I serve as senior author. Yuxiang Feng is first author on *Ill-conditioning in Dictionary-Based Dynamic-Equation Learning: A Systems Biology Case Study* (Feng, Mangan, **M. Jayadharan**†, arXiv:2603.11330, under revision), for which I am senior and corresponding author. Henry Chen is first author on *Improving Extrapolation in Neural ODEs via Monomial Augmentation and Symbolic Structure Recovery* (Chen, Mangan, **M. Jayadharan**†, in preparation), for which I am senior author. These projects show my approach to mentoring students toward intellectual ownership, first authorship, and independent research contributions.

Awards, Fellowships, and Honors

- Travel award to participate in Midwest Numerical Analysis Day, University of Notre Dame, 2026.
- Travel award to attend the ACDC Autumn School, *From Microscopic Dynamics to Continuum Models*, VU Amsterdam / Leiden University, 2025.
- Travel award and invitation to speak, Indian Institute of Space Science and Technology (IIST), 2024.
- Provost's Dissertation Completion Fellowship, University of Pittsburgh, 2021.
- Graduate Student Researcher (NSF support), University of Pittsburgh, 2018–2020.
- Arts and Sciences Graduate Fellowship, University of Pittsburgh, 2016–2017.

- UGC–CSIR NET and GATE in Mathematics, 2016.
- Summer Research Intern Fellowship, NNMCB, IISc Bengaluru, 2015.
- Merit Certificate for Academic Excellence in Mathematics, IISER Mohali, 2014.
- DST INSPIRE Scholarship, Government of India, 2011–2016.
- Top 0.1% merit certificate in Physics, national CBSE exam, 2011.

Technical Skills

- **Languages:** Python, C++, Julia, MATLAB, Fortran, SQL.
- **Numerical / SciML:** FEM, sparse optimization, neural ODEs, SINDy / DAE-FINDER family, PINNs, multiple shooting, QR orthogonalization, weak-form / variational methods.
- **HPC:** MPI, Slurm, domain decomposition; deal.II, FEniCS, FreeFem++.
- **ML / Data:** NumPy, SciPy, SymPy, PyTorch, scikit-learn, scientific Python workflows; data-driven modeling at scale.

Research Affiliations and Professional Service

- Affiliated researcher, National Institute for Theory and Mathematics in Biology (NITMB), an NSF–Simons-funded center for mathematical biology (current).
- Affiliated researcher, Trienens Institute for Sustainability and Energy, Northwestern University (current).
- Vice President, Association for Women in Mathematics (Pittsburgh chapter), 2020–2021.
- Officer, Graduate Student Organization, University of Pittsburgh, 2018–2020.

References

Available upon request.